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Roll No

MI-304 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Mining Surveying - I
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. a) State various types of chains used in linear measurements.
 b) Explain clearly the principle of chain surveying.
2. What is a well conditional triangle? Why is it necessary to use well-conditioned triangles?
3. The following bearing were observed while traversing with a compass.

Line	AB	BC	CD	DE
F.B.	N45°45'E	S83°05'E	N29°45'E	N35°12'W
B.B.	S46°10'W	N82°55'W	S29°10'W	S35°12'E

Mention which stations were affected by local attraction and determine the corrected bearings.

4. In table surveying. What are the various ways to orient a table at a station?
5. Define the following:(Any four)
 - i) True meridian
 - ii) Magnetic meridian
 - iii) Arbitrary meridian
 - iv) Magnetic declination
 - v) Reduced bearing
6. The following angles were observed in clockwise direction in an open traverse :
 $\angle ABC = 124^\circ 15'$; $\angle BCD = 156^\circ 30'$; $\angle CDE = 102^\circ 0'$; $\angle DEF = 95^\circ 15'$; $\angle EFG = 215^\circ 45'$
 Magnetic bearing of the line AB was $241^\circ 30'$. What would be the bearing of line FG?

OR

Discuss the method of surveying using Miner's dial.

7. The following offsets were taken from a chain line to a building.

Distance (cm)	0	6	12	18	24	36	48	60	72	84	95
Offsets (cm)	3.8	3.3	2.7	2.1	1.7	2.7	1.9	2.8	3.8	3.2	4.7

Calculate the area enclosed between the chain line the building and end offsets.

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OR

- a) What is local attraction? How is it detected?
- b) Define W.C.B. and R.B.

8. Answer any two of the following:

- a) Explain method of indirect ranging on a slopping ground, where the end stations are not inter-visible.
- b) Make a comparison between whole circle bearing system and quadrantal bearing system.
- c) What is meant by orientation in plane Table Survey? How is it done in the field.
- d) Describe common sources of Errors in dial surveying.
